

Xinyu Wang

CONTACT Yukawa Institute for Theoretical Physics, Kyoto University, Kyoto, Japan
Email: wang.xinyu.4v@kyoto-u.ac.jp

Website: wangxinyu4v-a11y.github.io
ORCID: 0009-0000-4370-2174

ACADEMIC APPOINTMENTS **Assistant Professor** September 2026–Present
Yukawa Institute for Theoretical Physics, Kyoto University, Japan

Postdoctoral Researcher 2023–2026
Department of Physics, University of Michigan, USA
Advisor: Prof. Xiaoming Mao

Postdoctoral Fellow 2021–2023
Department of Physics, Hong Kong University of Science and Technology, Hong Kong
Advisor: Prof. Rui Zhang

EDUCATION **Ph.D. in Engineering** March 2021
School of Civil Engineering, Southeast University, China Advisors: Prof. Jian Feng & Prof. Jianguo Cai Thesis: *Geometry and mechanics of origami and kirigami inspired metamaterials*

Visiting Ph.D. Student 2018–2020
Department of Physics and Astronomy, University of Pennsylvania, USA Advisors: Prof. Randall Kamien & Prof. Shu Yang

B.S. in Engineering 2011–2015
School of Civil Engineering, Southeast University, China

RESEARCH INTERESTS I develop theoretical and computational approaches to understand and design soft and architected materials, with a focus on nonlinear and topological mechanics, nonequilibrium behavior, and multiscale modeling. In close connection with experiments, my research investigates how geometry, topology, defects, and structural evolution give rise to robust, programmable, and adaptive mechanical responses.

PUBLICATIONS I have made primary contributions to 13 peer-reviewed papers as first author, co-first author, or second author (with my supervisor as first author) including *Physical Review X* (1), *Nature Communications* (1 published, 1 going through 3 round of peer review), *PNAS* (3) and *Composite Structures* (1).

Selected publications

See Google Scholar for citation.

† Equal contribution

1. X. Wang, S. Sarkar, S. Gonella, X. Mao, “Topological mechanical metamaterial for robust and ductile one-way fracturing”. *Nature Communications* 17, 2420 (2026).

2. J. Jiang[†], O.I. Akomolafe[†], X. Wang[†], Z. Asilehan, W. Tang, J. Zhang, Z. Chen, R. Wang, K. Ranabhat, R. Zhang, C. Peng. “Topology-driven collective dynamics of nematic colloidal entanglement”. *Proceedings of the National Academy of Sciences* (2024), 121(37): e2402395121.
3. X. Wang[†], J. Jiang[†], J. Chen[†], Z. Asilehan, W. Tang, C. Peng, R. Zhang, “Moiré effect enables versatile design of topological defects in nematic liquid crystals”. *Nature Communications* (2024), **15**, 1655 .
4. J. Jiang[†], X. Wang[†], Ol. Akomolafe[†], W. Tang, Z. Asilenhan, K. Ranabhat, R. Zhang, C. Peng, ”Collective transport and reconfigurable assembly of nematic colloids by light-driven cooperative molecular reorientations.” *Proceedings of the National Academy of Sciences* (2023), 120.16: e2221718120.
5. J. Jing[†], K. Ranabhat[†], X.Wang[†], H. Rich, R. Zhang, C. Peng, “Active transformation of topological structures in active nematic disclination networks”, *Proceedings of the National Academy of Sciences* (2022), 119(23), 2122226119.
6. X. Wang, S.D. Guest, R.D. Kamien, “Keeping It Together: Interleaved Kirigami Extension Assembly”, *Physical Review X* (2020), 10(1), 011013.

Others:

Papers:

1. S. Yi, H. Chen, X. Wang, M. Jiang, B. Li, Q. Wei, R. Zhang, “Line defects in nematic liquid crystals as charged superelastic rods with negative twist–stretch coupling”. preprint arXiv:2312.14735.
2. H. Liu, S. Sarkar, A. Arafat, X. Wang, E. Stanifer, S. Gonella, X. Mao, “ Stress control in non-ideal topological Maxwell lattices via geometry”, Under revision at *Physical Review B*.
3. X. Wang, R. Zhang. “ Design of shape morphing of liquid crystal elastomers through pre-shaping methods”. *Molecular Systems Design and Engineering* (2025), In press.
4. Q. Zhang, X. Yan, X. Wang, “Numerical modeling of gas flow and solid deformation in unconventional anisotropic shale”. *Computers and Geotechnics* (2025) Volume 187, November.
5. R. L., Dupont, Y. Xu, A. Borbora, X. Wang, F. Azadi, K. Havener, B. Lewis, W. Deng, B.W. Tan, S. Li, R. Zhang, Y. Yao, U. Manna, X. Wang, “Synergistic Adhesion and Shape Deformation in Nanowire-Structured Liquid Crystal Elastomers”. *Advanced Materials* (2025), p.2414695.
6. M. Tanaka[†], X. Wang[†], C. Mishra, J. Cai, J. Feng, R. Kamien, A. Yodh, “Ratchetlike motion of helical bilayers induced by boundary constraints”, *Physical Review E* (2022), 106(1), L012605.
7. X. Wang, J. Zhang, J. Cai, J. Feng, “An improved energy method for form-finding of mesh reflectors”, *Archive of Applied Mechanics* (2022), **92**, 2795–2812.
8. X. Wang, J. Cai, D. S. Lee, Y. Xu, J. Feng, “Numerical form finding of multi order tensegrity structures by grouping elements”, *Steel and Composite Structures* (2021), 41(2), 267-277.
9. Q. Zhang, X. Wang, J. Cai, R. Yang, J. Feng, “Prestress design for cable-strut structures by grouping elements”, *Engineering Structures* (2021), 244, 112010.

10. Q. Zhang, X. Wang, J. Cai, J. Feng, “Motion paths and mechanical behavior of origami-inspired tunable structures”, *Mechanics Research Communications* (2021), 26, 101872.
11. Q. Zhang, X. Wang, D. S. Lee, J. Cai, Z. Ren, J. Feng, “Development of kinetic origami canopy using Arc Miura folding patterns”, *Journal of Building Engineering* (2021), 43, 103116.
12. Q. Zhang, X. Wang, J. Cai, J. Zhang, Jian Feng, “Closed-Form Solutions for the Form-Finding of Regular Tensegrity Structures by Group Elements”, *Symmetry* (2020), 12, 374.
13. J. Cai, R. Yang, X. Wang, J. Feng, “Effect of initial imperfections of struts on the mechanical behavior of tensegrity structures”, *Composite Structures* (2019), 207, 871-876.
14. X. Wang, J. Cai, R. Yang, J. Feng, “Form-finding of deployable mesh reflectors using dynamic relaxation method”, *Acta Astronautica* (2018), 151, 380-388.
15. J. Cai, Y. Zhou, X. Wang, Y. Xu, J. Feng, “Dynamic analysis of a cylindrical boom based on Miura origami”, *Steel and Composite Structures* (2018), 28, 607-615.
16. J. Cai, X. Wang, X. Deng, J. Feng, “Form-finding method for multi-mode tensegrity structures using extend force density method by grouping elements”, *Composite Structures* (2018), 187, pp.1-9. (*Lead contribution; advisor listed first*)
17. J. Cai, X. Wang, R. Yang, J. Feng, “Mechanical behavior of tensegrity structures with High-mode imperfections”, *Mechanics Research Communications* (2018), 94, 58-63. (*Lead contribution; advisor listed first*)

Patents:

1. J. Cai (advisor), X. Wang, J. Feng, Y. Zhou. “Foldable cylindrical latticed shell structure”, Chinese Patent CN105756192A, Granted 2016.
2. J. Cai (advisor), X. Wang, J. Feng, Y. Zhou. “A kind of deployable cylindrical reticulated shell structure with six pieces of rigid plate folding unit”, Chinese Patent CN106284804A, Granted 2018.

TEACHING	Teaching Assistant for <i>Concrete Structures</i> (80 juniors) - <i>facilitated active learning sessions and integrated practical structural engineering examples</i>	2015-2017
	LEO Adjunct Lecturer, Physics Department, University of Michigan - <i>Teaching PHYSICS 250: Fundamental Physics for the Life Sciences II</i>	2026 Spring
MENTORING	Supervised undergraduate and master’s research - <i>projects on deployable and origami structures and first place award</i>	2016-2018
	Co-mentored Ph.D. students - <i>liquid crystal defect simulations, emphasizing modeling rigor, coding practices, and interpretation of complex data</i>	2021-2023
	REI undergraduates - <i>on shape-morphing vitrimers, nonlinear prestress, and frequency-analysis techniques</i>	2025

PRESS

Kirigami designs hold thousands of times their own weight
Penn Today. INVERSE. TechXplore. Cosmos.

PRESENTATIONS

“Topological Mechanical Metamaterial for Nonlinear Fracturing Control”, University of Bristol, June 2026. (Invited)
 “Geometry- and Topology-Programmed Functional Soft Materials Across Scales”, University of Massachusetts Amherst, March 2026. (Invited)
 “Topology-driven versatile design and collective dynamics in nematic liquid crystals”, Geometers of Soft Materials Symposium, University of Pennsylvania, March 2026. (Invited)
 “Topology and Geometry for Programmable Architected Materials Across Scales”, University of Nebraska Lincoln, March 2026. (Invited)
 “Applied Topology-driven Design for Structural and Soft Materials”, University of Birmingham, November 2025 (virtual). (Invited)
 “Geometry- and topology-programmed functional materials across scales”, City University of Hong Kong, Hong Kong, August 2025. (Invited)
 “Topological mechanical metamaterial for robust and ductile one-way fracturing”, Global Physics Summit, Anaheim, USA, March 2025.
 “Mechanical metamaterial from geometry and topology”, University of Science and Technology of China, Hefei, China, September 2024. (Invited)
 “Moiré effect enables versatile design of topological defects in nematic liquid crystals”, contributed oral talk, The 8th International soft matter conference (ISM), Raleigh, USA, 2024.
 “Fracturing and Controlled Cracking Path in Topological Maxwell Lattice”, APS March Meeting, Minneapolis, USA, March 2024.
 “Moiré effect enables versatile design of topological defects in nematic liquid crystals”, University of Pennsylvania, Nov 2023. (Invited)
 “Shape Selection of Pre-shaped Liquid Crystal Elastomer Strips”, The 26th Annual Conference of HKSTAM, Hong Kong, 2023.
 “Emerging Topological Structures in Nematic Moiré Patterns”, APS March Meeting, Las Vegas, USA, 2023.
 “Shape-selection of Pre-shaped Liquid Crystal Elastomers”, Soft Matter Day, Hong Kong University of Science and Technology, December 2022.
 “Active control of Periodic, Three-Dimensional Disclination Networks in Nematic Liquid Crystals”, APS March Meeting, Chicago, USA, March 2022.
 “Origami Based Cylindrical Mechanical Metamaterial”, In Proceedings of IASS Annual Symposia. International Association for Shell and Spatial Structures (IASS), MIT, July 2018.
 “A Deployable Origami Tube for Light-Shielding Space Craft” (poster), the Second International Workshop on Origami Engineering, Tianjin, China, November 2017.

GRANTS AND AWARDS

University of Michigan, Postdoc Travel Award	2024
HKUST, Research Travel Grant	2023
HKUST, Post-Doctoral Fellowship (PDF) Matching Fund	2021
Chinese Scholar Council Fellowship, Chinese government	2018-2019
Postgraduate Research & Practice Innovation Grant, Jiangsu Province (KYCX18.0105)	2018
Prestress Structure Scholarship, Southeast University	2018

ACADEMIC SERVICES**Reviewers**

Advanced Functional Materials, Journal of Applied Physics, Journal of Chemical Physics, Journal of Elasticity and others.

Outstanding Postdoctoral Fellow Award, University of Michigan

Executive roles and organizers

Communication Officer in APS FECS	2025-2027
Organization assistant of Soft Matter Day 2022, HKUST	2021-2022
Organization assistant of a joint symposium, HKUST	2021
Organization assistant of Deployable Structure Workshop, Southeast University	2016, 2018

**OUTREACH
ACTIVITIES**

Squishy Science Sunday, Global Physics Summit	2025
- <i>“Talk to a physics”</i> part	
Facilitator, Philly Materials Science and Engineering Day	2019, 2020
- <i>Showed kids origami and kirigami inspired designs and potential applications</i>	
Facilitator, summer project in Penn GRASP lab	2019
- <i>Taught high school students how to make kirigami motifs</i>	